Do Synthetic Personas Predict Real Conversion? A Blind, Leakage-Controlled Backtest Across Three Decision Framings

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Abstract

Synthetic personas are increasingly proposed as a cheap substitute for real demand validation, yet their ability to predict real-world conversion is seldom tested against ground truth. We build a blind, leakage-controlled backtest—five documented historical validation episodes, anonymized so an LLM cannot recall their outcomes—and have persona agents from a population-grounded dataset make purchase decisions, scoring against the real result. Across 200 personas per case we find naive one-shot persona simulation is anti-predictive: real successes receive lower simulated conversion than real failures (success−failure separation $\Delta = -0.46$). Crucially, this failure is framing-invariant: reformulating the decision from stated willingness to a longitudinal revealed-preference judgment collapses every response to a floor ($\Delta = 0$), and a graded-probability formulation restores resolution but not discrimination ($\Delta = -0.05$); the case ranking is identical across all three. The failure is also aggregation-invariant: real successes are dominated by failures not only in the mean but at every quantile including the top decile and the maximum, so no hidden enthusiast minority explains the real winners. Applying the method to two real pre-launch products reproduces the bias (predicting $\sim$14% paid conversion for a product whose team targets 5%). Yet the anti-correlation is consistent: a leave-one-out calibration on a few real outcomes recovers perfect directional prediction (5/5), showing the simulation carries real but inverted signal. We conclude synthetic-persona demand sims should not be used raw, but become predictive once calibrated against a handful of real outcomes—and we release the blind backtest as a reusable validity gate.

1. Introduction

AI coding lets founders ship faster than ever, yet few products sustain value; demand validation (interviews, fake-door tests, A/B) is the recommended hedge but is costly for solo builders. Synthetic personas promise to automate it, and recent work offers population-grounded persona datasets (NVIDIA, 2025) and multi-agent simulation engines (MiroFish contributors, 2025). The unanswered question is epistemic: can a persona simulation predict conversion that holds in the real world? We contribute (i) a blind, leakage-controlled backtest protocol isolating predictive signal from LLM memorization via case anonymization; (ii) a three-framing robustness study (stated, revealed, graded) showing the failure is not an artifact of one prompt; and (iii) a cautionary real-product application to two live pre-launch products; and (iv) a calibration result showing the consistent bias is correctable, recovering 5/5 leave-one-out directional prediction. The raw finding is negative, obtained without tuning, but the systematic bias proves usable once calibrated.

2. Related Work

Population-grounded persona datasets (NVIDIA, 2025) supply demographically representative personas at scale, and multi-agent engines (MiroFish contributors, 2025) release many LLM agents into a synthetic environment to forecast collective dynamics. Industry reports use synthetic customers for pricing and NPS. Prior work mostly evaluates plausibility; we instead test external predictive validity against known conversion outcomes under blind, leakage-controlled conditions—the novel combination being rigorous, baseline-relative backtesting of conversion.

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Table 1. Graded per-case conversion (200 personas, general). Successes A,B,E score at or below failures C,D. Ranking C>D>B>A>E is identical under all three framings.

Case	Method	GT	Graded
A	fake-door video/waitlist	1	0.153
B	concierge fake store	1	0.202
C	reformulate flagship	0	0.230
D	Wizard-of-Oz	0	0.180
E	landing-page pricing	1	0.126

Table 2. Separation Δ across three framings and two persona conditions. Every value is ≤ 0; no configuration is predictive. Revealed collapses to a floor (all responses 0).

Framing	General Δ	Target Δ
No-persona baseline	-0.233	
Stated (binary)	-0.462	-0.494
Revealed (binary)	0.000	0.000
Graded (continuous)	-0.045	-0.054

3. Method

Personas. We sample personas from a $\sim$1M-persona population-grounded Korean dataset (NVIDIA, 2025) (occupation, education, region, consumption narrative), rendered to a compact profile. A local 7B instruction-tuned LLM (Qwen Team, 2024) is prompted to become the persona.

Three decision framings. Stated: one-shot "would you pay?" (binary). Revealed: a longitudinal judgment—click, try, and still pay after 2–8 weeks once novelty fades and incumbents are considered (binary). Graded: a continuous probability of becoming a retained paying customer. A case's simulated conversion is the mean over its personas.

Persona conditions. General: random personas. Target: personas keyword-filtered to each product's target segment, fixed a priori from the product description (never the outcome).

Baseline and leakage control. A no-persona baseline asks the LLM for the conversion rate directly. Each case is anonymized (brand, company, era removed) so the model cannot recall the historical result; ground truth is revealed only for scoring. We report directional accuracy and separation $\Delta = \overline{\text{sim}}_{\text{success}} - \overline{\text{sim}}_{\text{failure}}$ ($\Delta > 0$ predictive, $\Delta < 0$ anti-predictive).

4. Results

Table 1 gives per-case simulated conversion under the graded framing; real successes (A,B,E) sit below real failures (C,D). Table 2 shows this is framing-invariant: no framing or persona condition yields $\Delta > 0$.

Directional accuracy never exceeds the no-persona baseline (0.40). The stated framing over-predicts appealing-but-failed cases; the revealed framing is so strict that every persona declines (a degenerate floor); the graded framing restores resolution but leaves failures ranked above successes. The invariance of the ranking across framings, persona conditions, and against the baseline indicates a structural, not prompt-specific, failure.

Aggregation robustness. One might suspect the mean hides an intense niche minority for the real successes. It does not: recomputing per-case distributions, the failures dominate the successes on the high-intensity fraction, the top-decile mean ($\Delta = -0.21$), the standard deviation, and the maximum ($\Delta = -0.32$). No moment of the persona distribution is predictive, closing the "wrong statistic" objection.

Real-product application. We apply the graded method to two live pre-launch products (anonymized): P1, a real-time civic-debate app with a paid subscription tier, and P2, an open-source multi-agent tool for solo founders. Predicted paid conversion is 0.14 (P1) and 0.10 (P2), essentially unchanged under target conditioning. P1's own team targets a 5% premium conversion; the simulation over-predicts by $\sim 3\times$, reproducing the abstract-appeal bias on real inputs.

Calibration recovers prediction. The failure is systematic, not random: across cycles the real successes sit consistently below the real failures. We exploit this with leave-one-out calibration—for each held-out case, a one-feature decision stump is fit on the other four and used to predict it. Using stated conversion or the two-month retention level, LOO directional accuracy is 5/5 (versus 0.40 raw); the averaged signal also reaches 5/5. The simulation thus encodes genuine but inverted demand signal that a few real outcomes suffice to decode. The caveat is $N = 5$: perfect LOO on five cleanly separated cases is encouraging but underpowered, and pre-launch products without real labels (P1, P2) still require a small real anchor (e.g., a fake-door test) to calibrate.

5. Discussion: Why It Fails

(1) Abstract-appeal bias. One-shot willingness elicits stated appeal, not revealed conversion; broadly-

appealing-but-failed products (a sweeter drink, effortless dictation) score high. (2) Missing temporal/relational dynamics. The real failures hinge on brand attachment and sustained-use fatigue—invisible to a single decision; forcing a longitudinal binary instead floors everything. (3) Counterintuitive niche demand. Real successes had low general appeal but strong niche demand, which target filtering and graded probabilities still underrate. Because the same ranking survives every reformulation, the limitation is intrinsic to persona willingness, not to a particular prompt. Critically, however, the bias is consistent enough that light calibration against real outcomes (Section 4) converts it into a usable predictor: the practical recommendation is to treat persona simulations as a calibrated feature, never a raw estimate.

6. Limitations

Five backtest cases yield low statistical power; we frame results as a mechanism-focused, framing-robust case study, not a definitive effect size. We use one 7B model, one (Korean) persona market, and keyword target definitions; larger models, multi-turn deliberation, and revealed-behavior traces are untested and are exactly what our diagnosis motivates. The real-product predictions have no ground truth and are reported only to illustrate the bias.

Impact Statement

This work cautions against substituting synthetic-persona simulations for real user evidence in product decisions; over-trust could bias which products get built and entrench dataset demographics. By reporting a rigorous, framing-robust negative result we aim to temper deployment claims. No human subjects or private data are involved; product references are anonymized.

Software and Data

Code, prompts, anonymized case corpus, and per-persona logs for all three framings are available at an anonymized repository and will be de-anonymized upon acceptance.

References

MiroFish contributors. MiroFish: A simple and universal swarm intelligence engine. https://github.com/666ghj/MiroFish, 2025. Multi-agent prediction simulation engine.

NVIDIA. Nemotron-personas-korea. https://huggingface.co/datasets/nvidia/Nemotron-Personas-Korea, 2025. Population-grounded synthetic persona dataset.

Qwen Team. Qwen2.5: A party of foundation models. https://huggingface.co/Qwen/Qwen2.5-7B-Instruct, 2024.

A. Anonymized Case Corpus

A device-syncing tool via demo-video waitlist (real: success); B online shoe store via concierge fake store (success); C reformulated flagship drink that won blind taste tests, replacing the original for loyal customers (failure); D voice dictation assessed by Wizard-of-Oz for sustained use (low demand); E social-post scheduler via landing-page pricing (success). Ground truth is revealed only for scoring.

B. Framings, Prompts, and Targets

Stated elicits would_pay; revealed elicits retained_pay after a click→try→retain funnel with incumbent attachment; graded elicits pay_prob$\in [0, 1]$. Targets were fixed a priori by keyword (e.g., E: marketing/creator/self-employed terms) matched against occupation and persona-narrative fields. Matched target counts (8000-pool): A 3062, B 711, C 2034, D 2834, E 1023. Full prompts and keyword lists are in the repository.

C. Per-Framing Logs

Per-persona decisions and free-text reasons are in results_{stated,revealed,graded}_{general,target}.json and, for real products, results_products_graded_*.json.